ROM is, due to the nature of how it stores information. This is why RAM is often used to shadow the BIOS ROM to improve performance when executing BIOS code.

### SRAM

Static RAM; a type of RAM that holds its data without external refresh, for as long as power is supplied to the circuit. This is contrasted to dynamic RAM (DRAM), which must be refreshed many times per second in order to hold its data contents. SRAMs are used for specific applications within the PC, where their strengths outweigh their weaknesses compared to DRAM. SRAMs don't require external refresh circuitry or other work in order for them to keep their data intact. And SRAM is faster than DRAM.

In contrast, SRAMs have the following weaknesses, compared to DRAMs: SRAM is several times more expensive than DRAM. And SRAMs take up much more space than DRAMs. 32 MB of SRAM would be prohibitively large and expensive, which is why DRAM is used for system memory. SRAMs are used for level 1 cache and level 2 cache memory, for which it is perfectly suited—cache memory needs to be very fast, and not very large.

### DRAM

Dynamic RAM; a type of RAM that only holds its data if it is continuously accessed by special logic called a refresh circuit. This circuitry reads the contents of each memory cell several hundreds of times a second, whether or not the memory cell is being used at that time. Due to the way in which the cells are constructed, the reading action itself refreshes the contents of the memory. If this is not done regularly, the DRAM will lose its contents, even if it continues to have power supplied to it. This refreshing action is why the memory is called dynamic.

### Asynchronous And Synchronous DRAM

Conventional DRAM is said to be asynchronous. This refers to the fact that the memory is not synchronised to the system clock. A memory access is begun, and a certain period of time later, the